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Data Visualization

VIS101 | Suggested sequence · Semester 4 | Student-authored study guide

How to use these notes

Choose visual encodings, design readable charts, and communicate comparisons and uncertainty honestly. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Question, data, and chart choice

Start from a comparison, trend, distribution, relationship, or composition question. Match chart form to data: line for ordered trends, scatter for relationships, histogram for distributions, bars for category comparisons.

Worked example

```
Show monthly counts in chronological order with a line; do not connect unordered category
names with a line.
```

Check your understanding

- Which chart reveals a distribution?
- Why should time be sorted?

MODULE 02 · Perception, labels, and accessibility

Position and length are often easier to compare than area or color. Use readable labels, units, meaningful scales, and color palettes that remain interpretable without hue alone. Provide alt text or a text summary.

Worked example

```
Use a zero baseline for ordinary bar charts and label a truncated line-chart axis clearly.
```

Check your understanding

- Why can a misleading axis distort a comparison?
- Name one accessible encoding besides color.

MODULE 03 · Uncertainty and narrative

Distinguish observed data from estimates. Show intervals or distributions when uncertainty matters and disclose denominators and missing data. A title should state the finding without overstating causality.

Worked example

```
Describe “Group A had a higher sample mean” rather than “Group A causes higher outcomes” for
observational data.
```

Check your understanding

- When should confidence intervals be shown?
- Why report a denominator?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.